

DayNight Significance Workflow

Introduction

This presentation summarizes the workflow and outputs of the DayNight significance analysis for the SOLAR project.

- This deck is auto-generated from workflow outputs.
- This deck is scoped to the **Truncated** folder for the **SolarEnergy** reconstruction algorithm.

Config aliases:

- hd_1x2x6_centralAPA: HD Central
- hd_1x2x6_lateralAPA: HD Lateral
- vd_1x8x14_3view_30deg_nominal: VD Top
- vd_1x8x14_3view_30deg_shielded: VD Bottom Shielded

Workflow

- config: list of detector configs
- folder: **Truncated**
- analysis: DayNight
- exposure: default **30 years**
- threshold in 12DayNight.py: default 8.0 MeV
- optional cuts override: nhits, ophits, adjcls
- best-curve reference in 0ZBestSigmas.py: Smoothed or Raw

Workflow Outputs

- Fiducial optimization: `data/solar/fiducial/truncated/BestFiducials.json`
- Best cut summaries (JSON): `data/analysis/best-sigma-json/daynight/truncated`
- Backward-compatible local fallback: `data/analysis/daynight-json/truncated`
- Significance scans (PNFS outputs):
`/pnfs/ciemat.es/data/neutrinos/DUNE/SOLAR/DAYNIGHT/truncated`
- Figures: `images/analysis/day-night/truncated`

Day-Night Discovery Statistic

- `src/analysis/12DayNight.py` forms the baseline spectrum as background plus the day signal, using $B_i = B_i^{raw}/2 + S_i^{day}$ and the corresponding smoothed components above threshold.
- The discovery spectrum is the day-night difference, $\Delta S_i = S_i^{night} - S_i^{day}$, evaluated in the thresholded energy region.
- Per-bin discovery proxies are computed with `evaluate_significance(..., type="gaussian")` in `lib/lib_sigma.py`, then combined into the global curve as:

$$Z_{global} = \sqrt{\sum_i Z_i^2}.$$

Day-Night Discovery Statistic Details

- [src/analysis/12DayNight.py](#) stores both the plain Gaussian curve and an alternate version with background-statistical uncertainty included; this workflow does not perform a likelihood or chi-square fit.
- Smoothing is only used to build alternate component spectra: each component is smoothed separately, and the threshold slice keeps unsmoothed bins below threshold while replacing bins above threshold with their smoothed values.

Histogram Smoothing Math I

- Linear smoothing model used per histogram bin:

$$\tilde{h}_i = \sum_j K_{ij} h_j$$

- Integral-preserving normalization applied after smoothing:

$$\tilde{h}_i \leftarrow \tilde{h}_i \cdot \frac{\sum_j h_j}{\sum_j \tilde{h}_j}$$

Histogram Smoothing Math II

- Threshold-slice smoothing used for DayNight threshold region:

$$h_i^{\text{out}} = \begin{cases} h_i, & i < i_{\text{thr}} \\ \tilde{h}_i, & i \geq i_{\text{thr}} \end{cases}$$

- Variance propagation through the same linear operator:

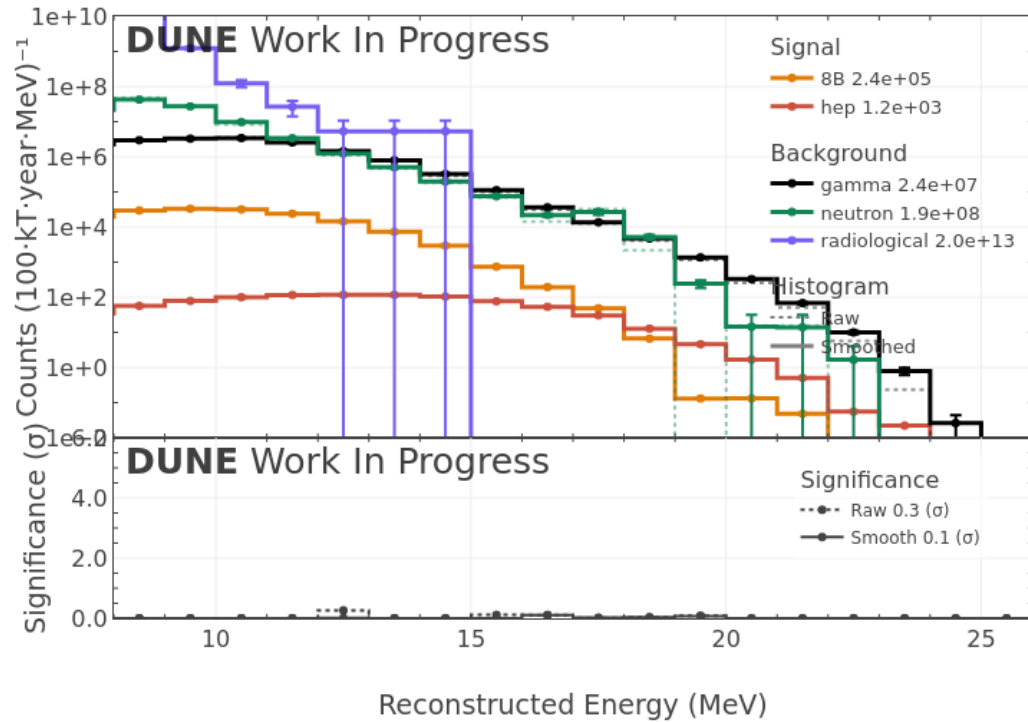
$$v^{\text{out}} = (K \odot K) v, \quad \sigma_i^{\text{out}} = \sqrt{v_i^{\text{out}}}$$

Fiducialization

HD Central

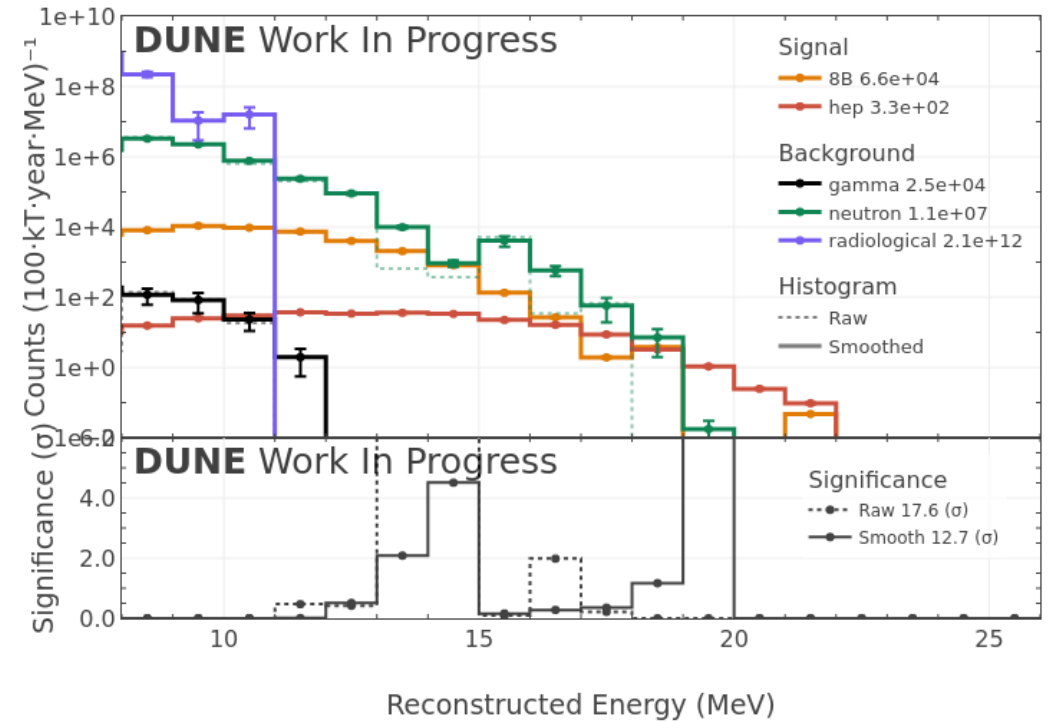
No Fiducial

DayNight SolarEnergy Significance
Fiducial: X=0cm, Y=0cm, Z=0cm



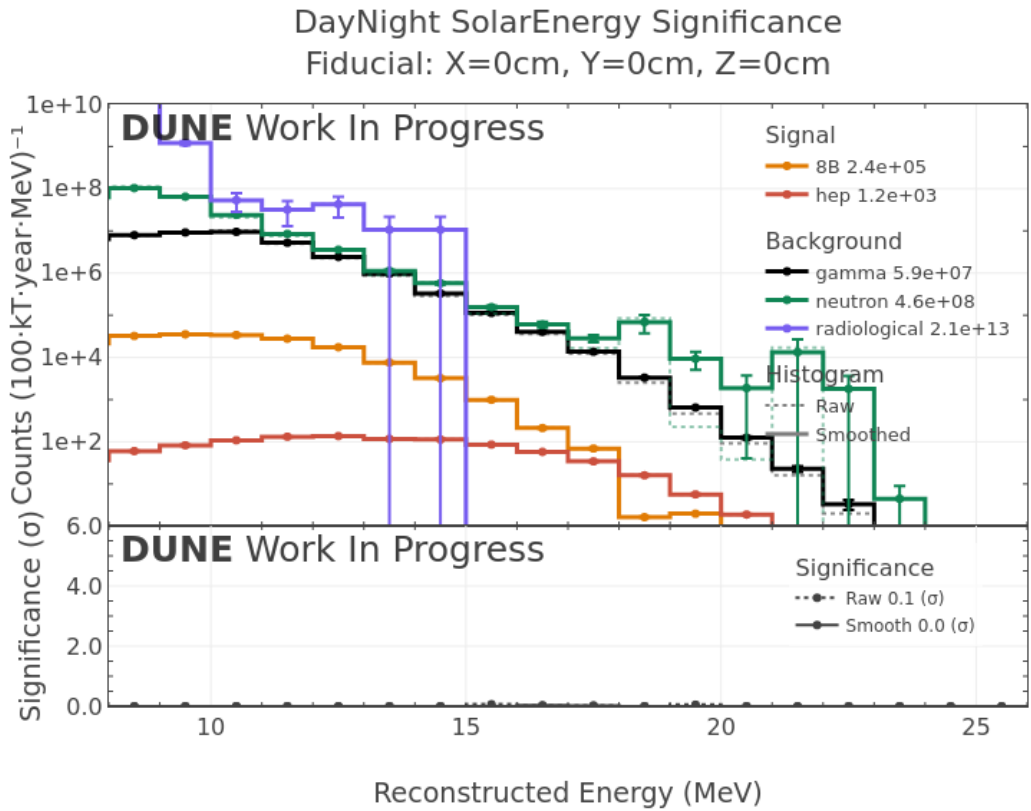
Best Fiducial

DayNight SolarEnergy Significance
Fiducial: X=20cm, Y=280cm, Z=280cm

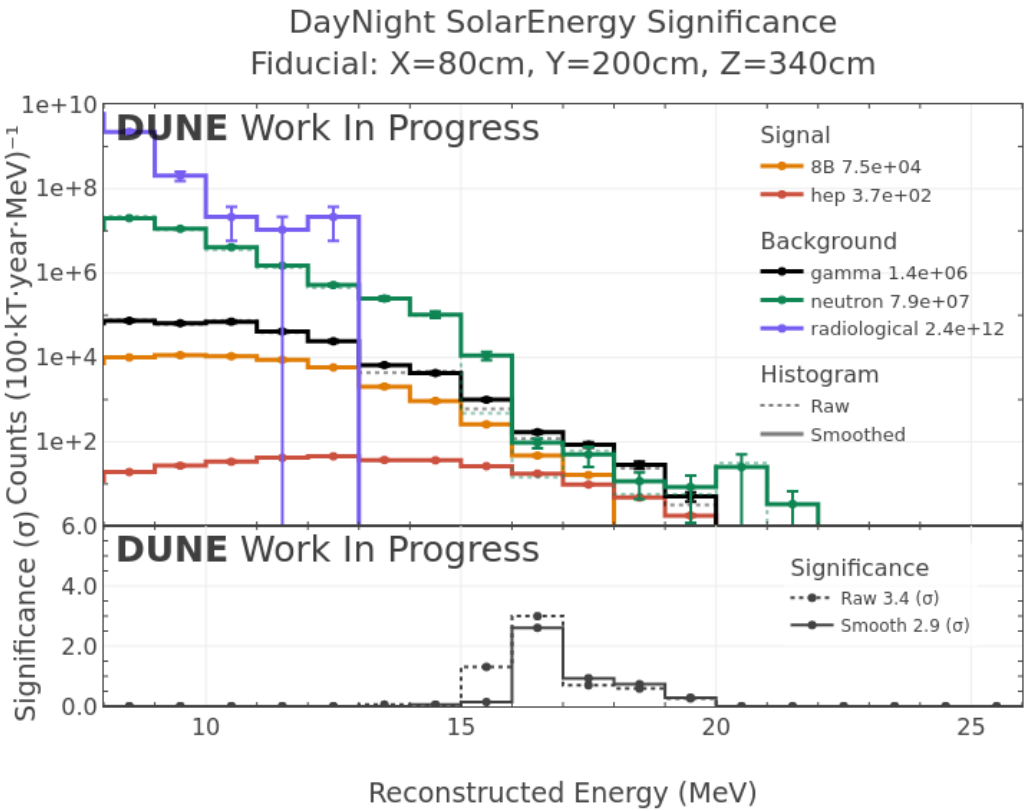


HD Lateral

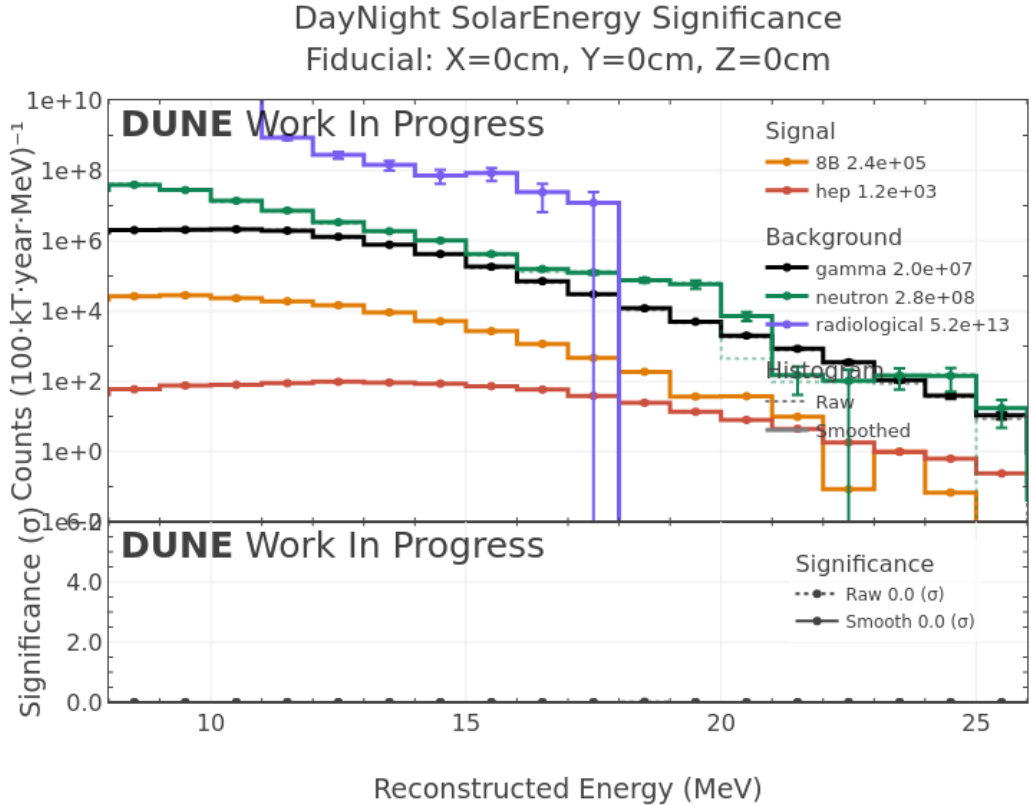
No Fiducial



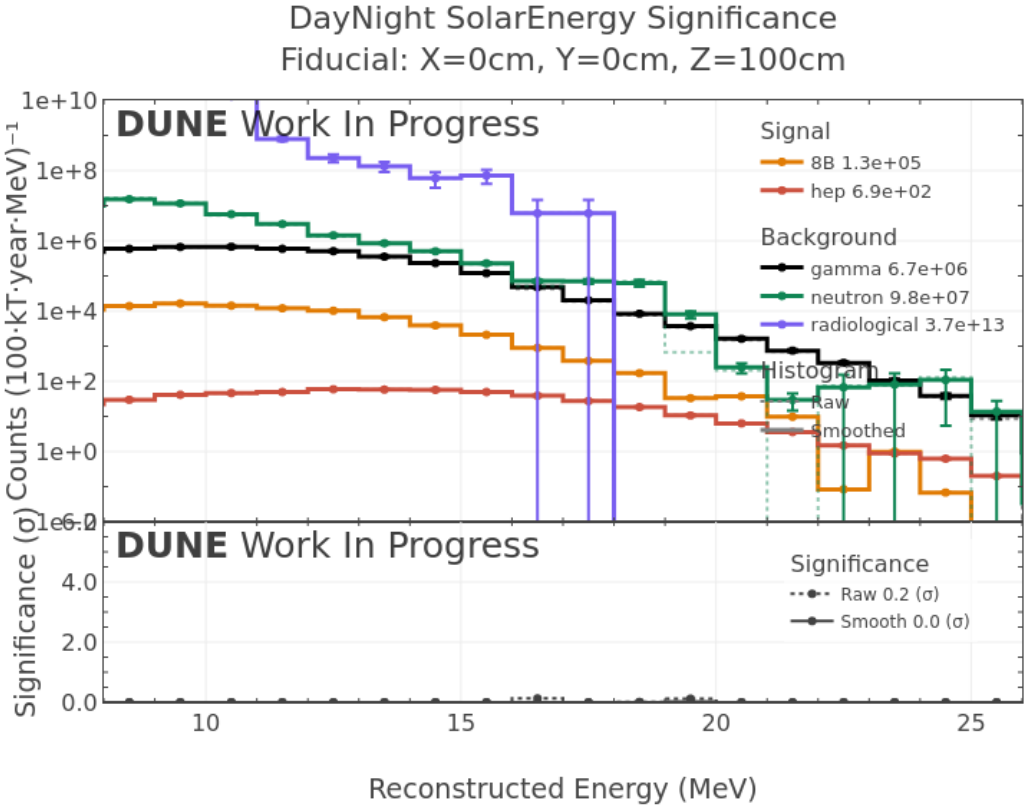
Best Fiducial



No Fiducial

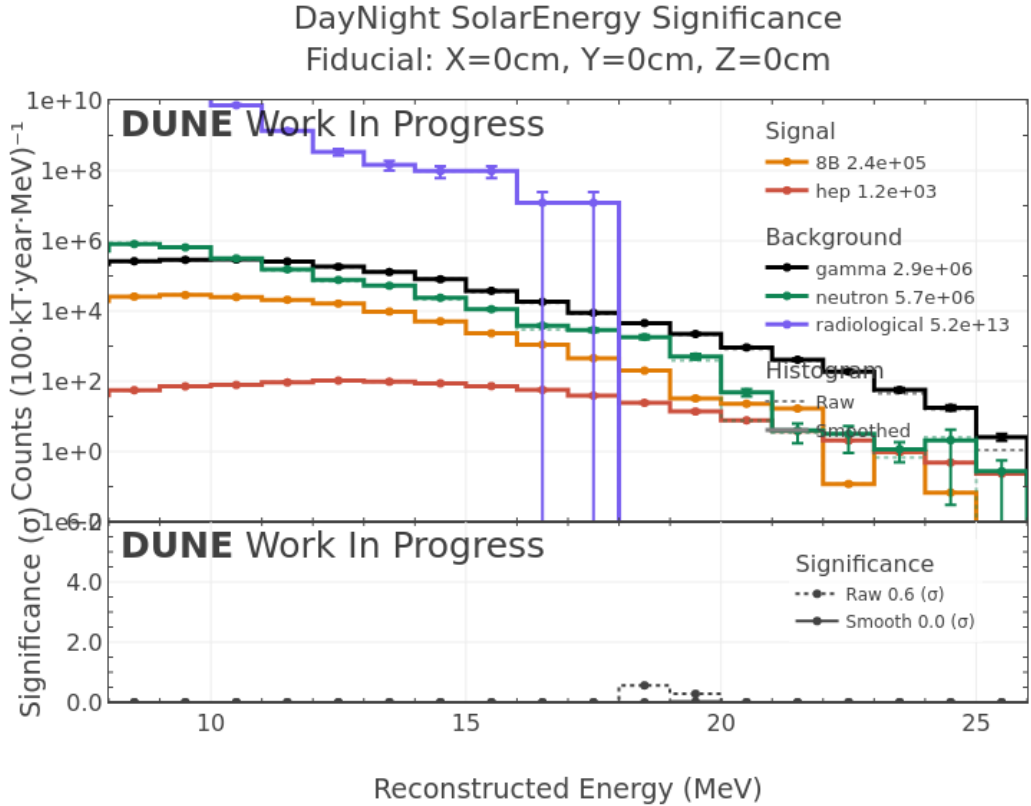


Best Fiducial

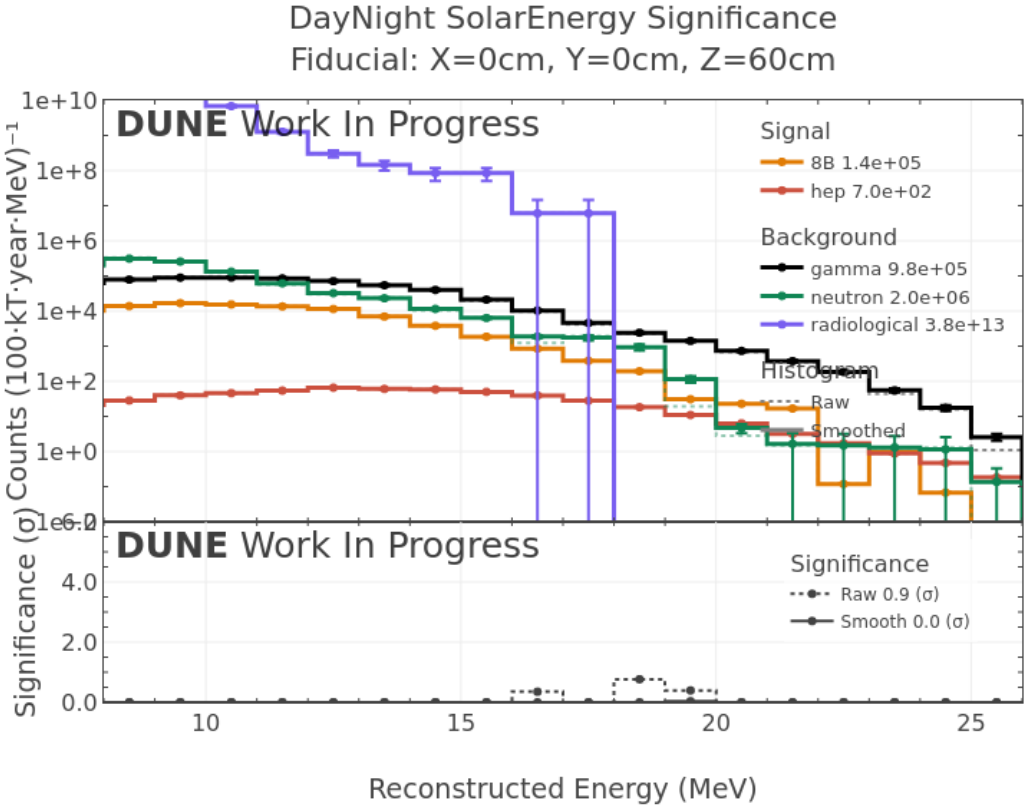


VD Bottom Shielded

No Fiducial



Best Fiducial



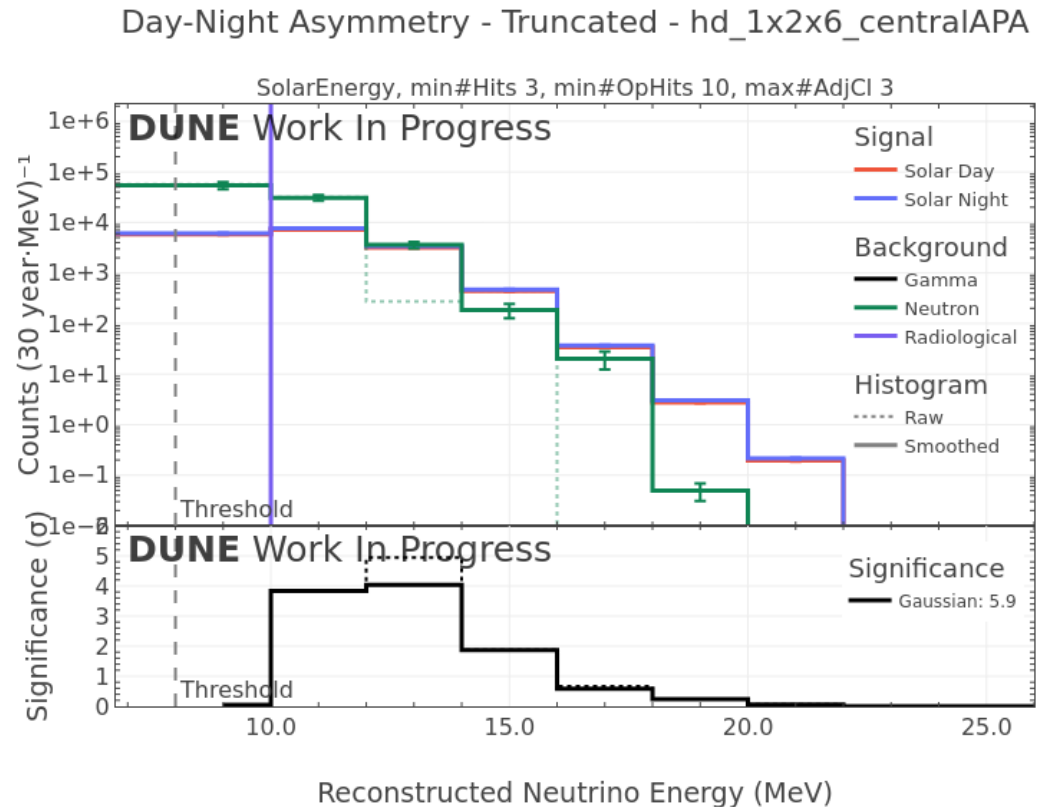
Fiducial Optimization Summary

Config	Fiducial X	Fiducial Y	Fiducial Z	Before Fiducialization	After Fiducialization
HD Central	20	280	280	0.131	8.268
HD Lateral	80	200	340	0.040	2.293
VD Top	0	0	100	0.004	0.024
VD Bottom Shielded	0	0	60	0.020	0.036

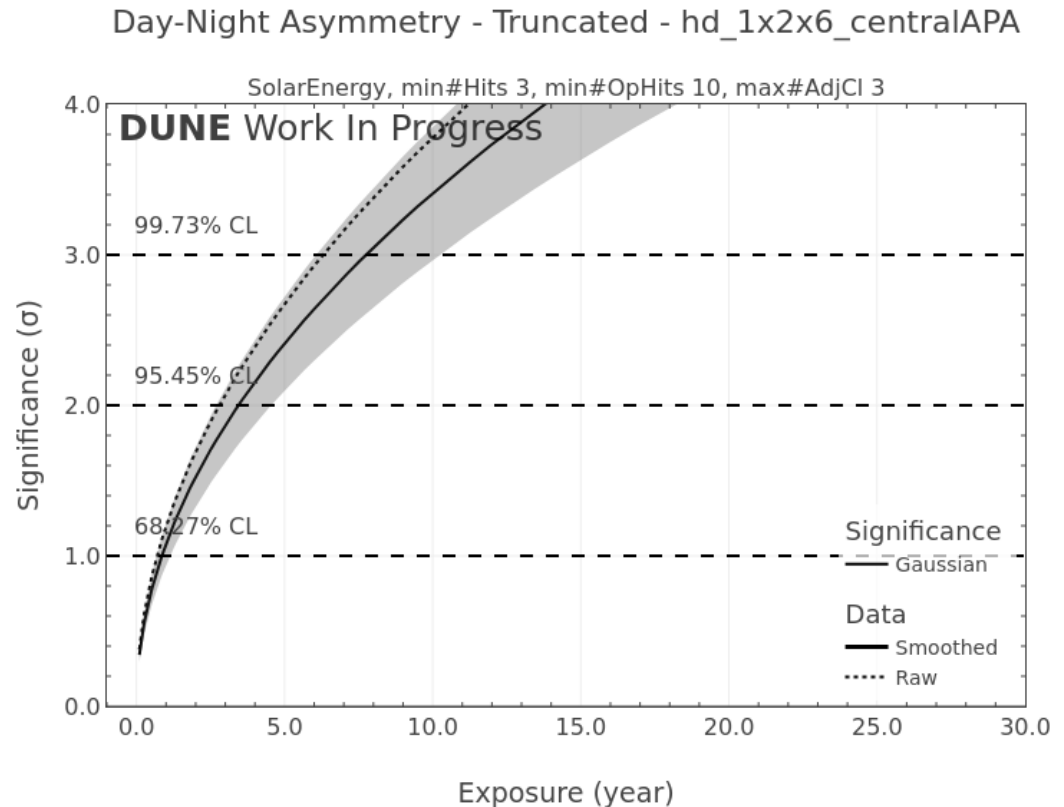
DayNight Results

HD Central

Significance

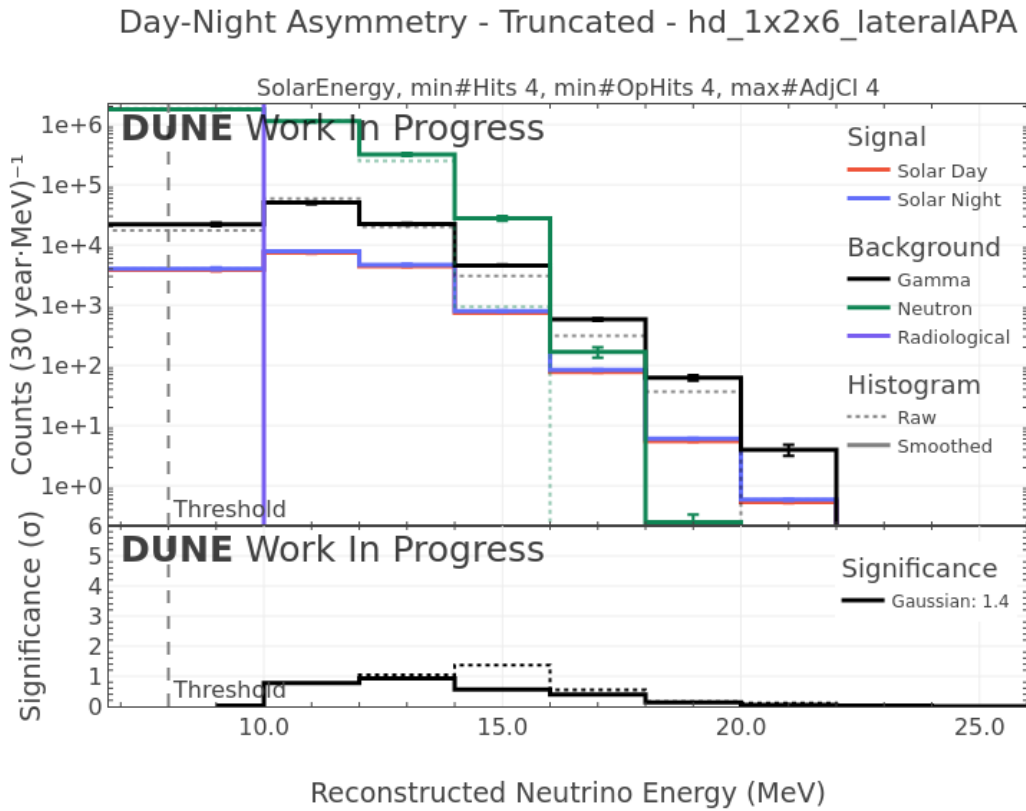


Exposure

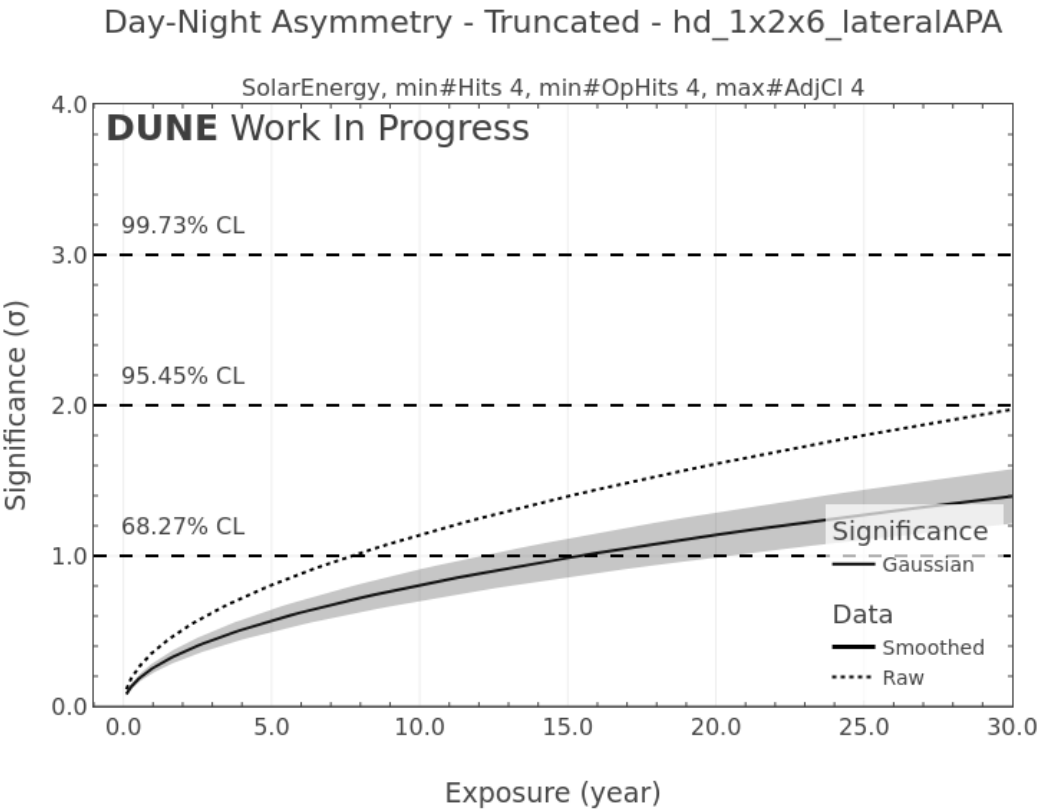


HD Lateral

Significance

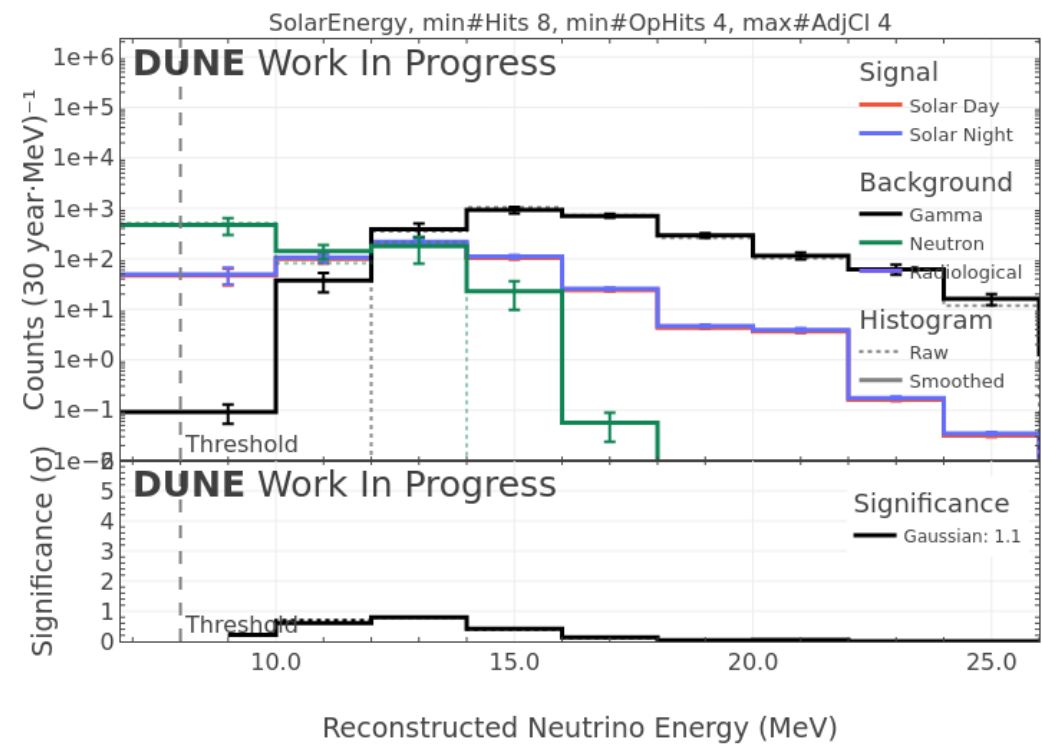


Exposure



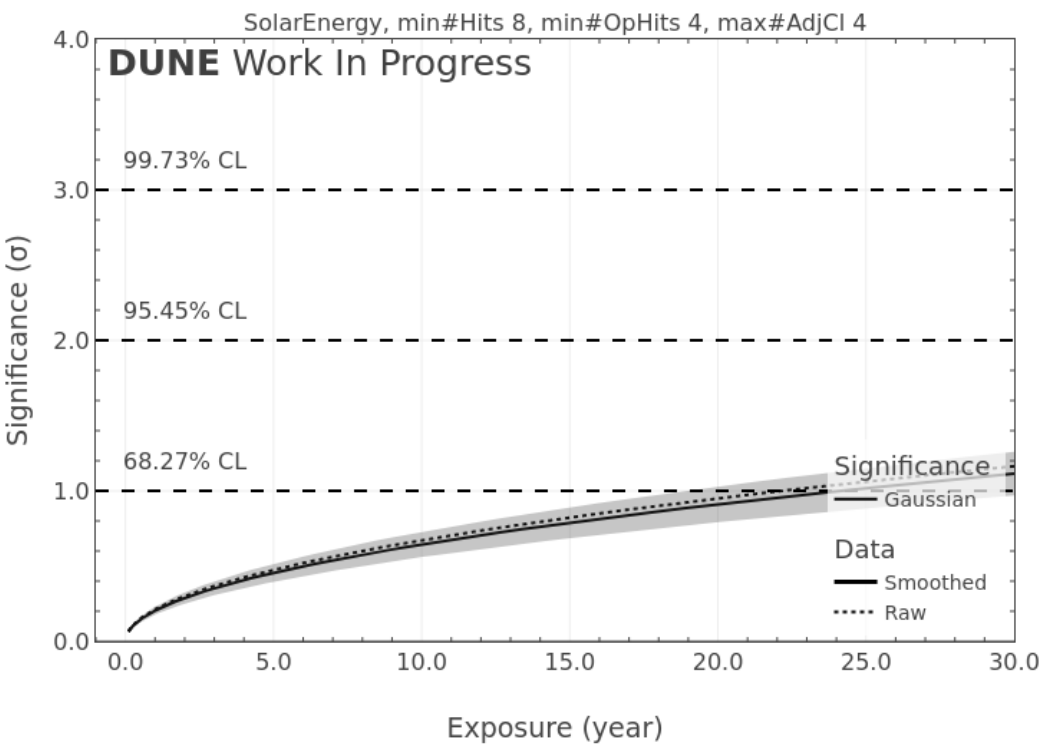
Significance

Day-Night Asymmetry - Truncated - vd_1x8x14_3view_30deg_nominal



Exposure

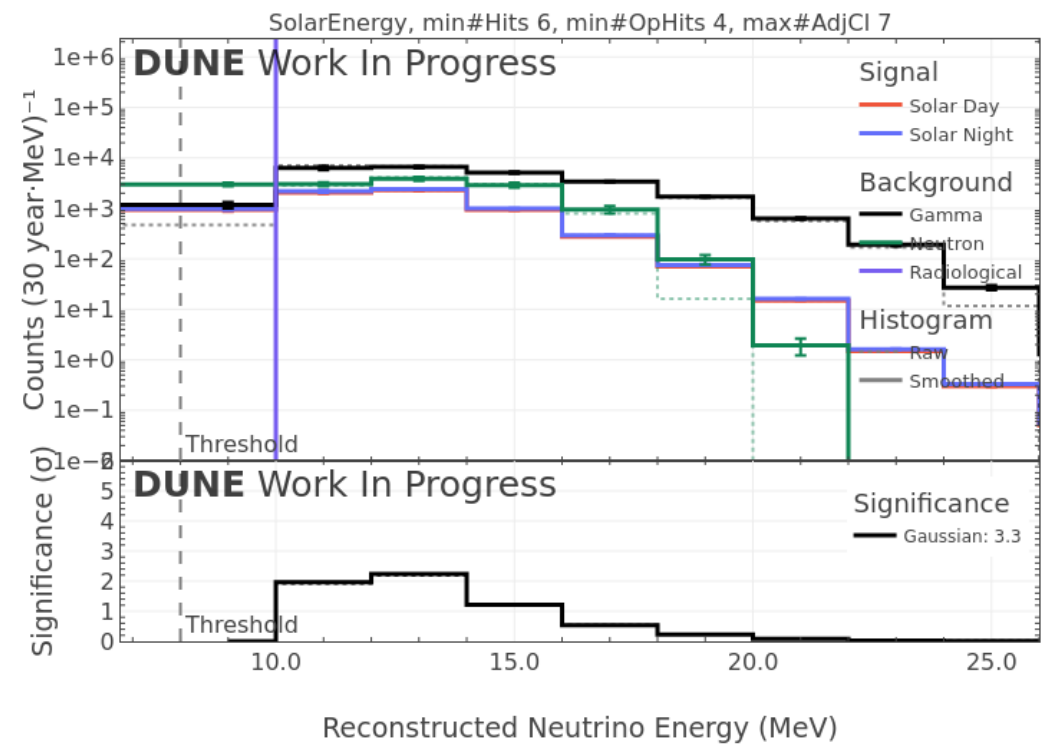
Day-Night Asymmetry - Truncated - vd_1x8x14_3view_30deg_nominal



VD Bottom Shielded

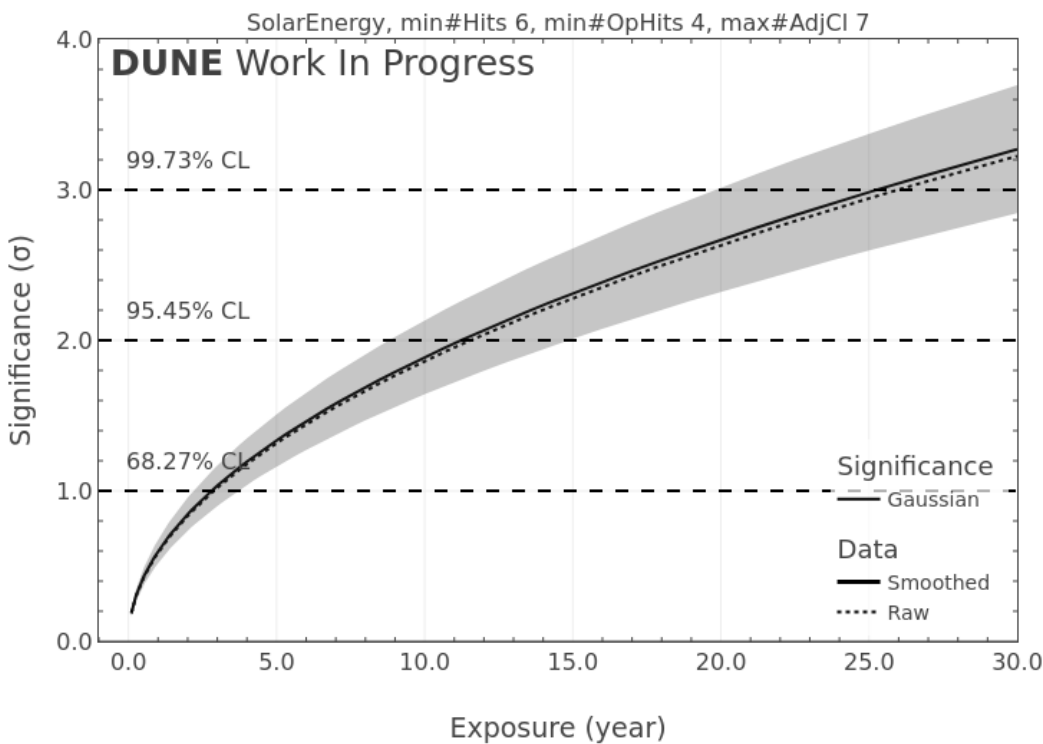
Significance

Day-Night Asymmetry - Truncated - vd_1x8x14_3view_30deg_shielded



Exposure

Day-Night Asymmetry - Truncated - vd_1x8x14_3view_30deg_shielded



Best DayNight Cuts by Config

Config	NHits	OpHits	AdjCI	Significance
HD Central	3	10	3	5.903
HD Lateral	4	4	4	1.394
VD Top	8	4	4	1.114
VD Bottom Shielded	6	4	7	3.269

Coverage and Notes

- Config coverage in best-cut JSON outputs:
 - nominal: 5
 - reduced: 4
 - truncated: 4
- Table values are read from workflow-generated JSON at generation time.
- Re-run script to refresh this folder after each workflow run:
- `/usr/bin/python3 scripts/generate_daynight_presentation.py --folder truncated`